

		m
12	C	For a satellite that is in orbit, the gravitational force provides for the centripetal force, $G \frac{Mm}{r^2} = m \frac{v^2}{r}$ $mv^2 = G \frac{Mm}{r}$ The initial energy is the total energy of the satellite in orbit at radius R $TE_{\text{initial}} = KE + GPE = \frac{1}{2}mv^2 - \frac{GMm}{R} = \frac{GMm}{2R} - \frac{GMm}{R} = -\frac{1}{2} \frac{GMm}{R}$ The final energy is the total energy of the satellite at radius $2R$ is

		$TE_{\text{final}} = -\frac{1}{2} \frac{GMm}{2R}$ Energy needed = final energy – initial energy $= -\frac{1}{2} \frac{GMm}{2R} - \left(-\frac{1}{2} \frac{GMm}{R} \right) = \frac{GMm}{2} \left(\frac{1}{R} - \frac{1}{2R} \right)$
10	A	The frictional force is the restoring force